Housing Disparity

Is Housing Still Affordable? A Data Science Study of Income vs. Housing Costs

Author: Bryan Bucknell
Discipline: Data Science / Applied Statistics
Date: April 21, 2026

Abstract

This study examines trends in housing affordability in Canada over the period 2012–2023 by analyzing the relationship between income, housing costs, and inflation. Using national-level data, variables were standardized and adjusted for inflation to enable meaningful longitudinal comparison. Exploratory data analysis and statistical modeling were employed to assess patterns, relationships, and the extent to which housing costs have diverged from income over time.

The results indicate a substantial and sustained increase in housing costs, which rose by over 50% during the study period. In contrast, nominal income exhibited only modest growth, while real income declined when adjusted for inflation. Comparative analysis shows that housing costs increased at a significantly faster rate than both income and overall price levels, leading to a measurable increase in housing cost burden. Visual and statistical analyses further reveal a widening gap between income and housing costs, with evidence of non-linear growth in housing prices.

These findings suggest that housing affordability in Canada has deteriorated in a meaningful and persistent way, driven primarily by disproportionate increases in housing costs rather than corresponding gains in income. The results are consistent with the hypothesis that structural factors, rather than general economic growth alone, are contributing to affordability challenges.

While the analysis is limited by its use of aggregate data and observational methods, the consistency and magnitude of the observed trends provide strong evidence of a systemic imbalance. The study highlights the importance of integrated affordability metrics and supports the need for policy approaches that address both housing supply and income dynamics.

Background

In recent years more and more people have become worried about the rising cost of housing. ISurvey research conducted by the **Broadbent Institute** in 2023 found that **95% of Canadians** agree that rising rental costs and the lack of affordable housing constitute a serious problem with 57% reporting spending over 30% of income on rent or mortgage and 25% of renters spending over 50% of income on rent, well above commonly accepted thresholds Broadbent Institute

Housing, a necessity of life, has become such a strain that according to survey data from Abacus Data 57% of Canadians fear losing their homes if their financial situation changes. This concern is especially high amongst younger adults and lower income households.. Abacus Data

Housing stability is modeled as a function of income adequacy, housing cost burden, tenure security, household composition, and health-related constraints.

The study aims to examine shelter to cost income thresholds as defined by Statistics Canada and the Canada Mortgage and Housing Corporation.

Research Question

Are housing costs increasing at a faster rate than average household income in Canada?

Related questions to be examined:

How has the shelter-cost-to-income ratio changed over time?

Do trends in housing costs relative to income differ across income groups?

Are renters more affected than homeowners by changes in housing costs relative to income?

4. Hypotheses

This study examines whether housing costs in Canada have increased at a faster rate than average household income, and whether this divergence is associated with greater housing instability.

Housing costs in Canada have increased at a significantly faster rate than average household income over time.

.All hypotheses are tested using Canadian household-level or aggregate data, with statistical significance evaluated at conventional confidence levels.

Data Description

Income:
Average and median market, total and after-tax income of individuals by selected demographic characteristics
https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1110009101

1. Average and Median Market, Total and After-Tax Income of Individuals by Selected Demographic Characteristics

This dataset, **Table 11-10-0091-01**, provides annual estimates of individual income in Canada, including **average and median values** for multiple income concepts: **market income**, **total income**, and **after-tax income**. The table allows disaggregation by selected **demographic characteristics** (e.g., age, sex, education) and geographic units such as Canada, regions, and provinces or territories. It is part of the *Canadian Income Survey* and covers a multi-year reference period (2012–2023) with annual frequency.

- **Market income** refers to income received from employment, investments, and other sources before taxes and transfers.

- **Total income** includes market income plus **government transfers** (e.g., benefits).

- **After-tax income** is total income **after deduction of income taxes** and is commonly used to assess disposable income.
  Both **average** (mean) and **median** (midpoint) statistics are reported to characterize central tendencies and reduce skewness from high-income values. This table is useful for analysing income distribution, demographic income disparities, and temporal trends in income levels across population subgroups.

Housing costs

https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1110022401
Household spending by household type

Household Spending — Housing Costs and Total Household Expenditures by Household Type

The second dataset, **Table 11-10-0224-01**, presents **annual household spending data** by **household type** in Canada, compiled through the *Survey of Household Spending (SHS)*. It includes detailed expenditures on goods and services across **categories such as shelter (housing costs), food, transportation, clothing and other consumption items**, and total household spending. The table's coverage spans from 2010 to 2023 and can be customized by reference year.

- **Housing costs** are captured under the shelter category and may include rent, mortgage payments, utilities, and related expenses.

- **Household type** refers to classifications such as **single-person households, couples with/without children, and other family types**.
  The SHS data is used to understand **consumption patterns** across different household structures and to examine how spending on housing and other necessities varies with type and over time. Average spending figures reflect typical annual expenditures per household, allowing comparisons of economic burden and cost-of-living pressure among demographic groups. Statistics Canada+1

Data Cleaning & Preprocessing

To ensure analytical validity and comparability across datasets, a structured data cleaning and preprocessing pipeline was implemented prior to analysis. The primary objective was to harmonize income and housing cost data across time, remove inconsistencies, and construct variables suitable for longitudinal comparison.

Data Integration and Structure Alignment

Data were sourced from multiple tables provided by Statistics Canada, including datasets on individual income and household expenditures. As these datasets differed in structure, variables were standardized prior to merging.

Temporal alignment was achieved by extracting the reference year (`REF_DATE`) and ensuring consistency in date formats. All datasets were aggregated to the annual level to allow for direct comparison. A left join was performed on the year variable to preserve the integrity of the primary time series while incorporating auxiliary variables.

Rationale:
Combining datasets from different sources introduces structural inconsistencies. Standardizing temporal units and variable naming ensures that comparisons reflect real-world trends rather than artifacts of data formatting.

Handling Missing and Inconsistent Values

Missing values were assessed using completeness checks across all variables. Observations with missing key variables (income or housing costs) were excluded when imputation was not appropriate. For non-critical variables, missing values were retained where they did not affect the primary analysis.

Duplicate entries—particularly those arising from repeated categorical breakdowns (e.g., demographic subgroups)—were removed by filtering for aggregate categories (e.g., “All persons” or equivalent total population measures).

Rationale:
Excluding incomplete observations prevents distortion of trend estimates, while selective retention avoids unnecessary data loss. Filtering to aggregate categories ensures that results reflect population-level trends rather than subgroup-specific variation.

Inflation Adjustment and Real Value Conversion

All monetary variables were converted to constant dollars using a consumer price index (CPI) adjustment. This ensured that both income and housing costs were expressed in real terms (e.g., 2023 dollars).

Rationale:
Nominal values are not directly comparable across time due to inflation. Converting to real values isolates true changes in purchasing power and cost burden, which is essential for evaluating affordability trends.

Variable Construction

Key analytical variables were derived as follows:

- **Housing Cost Burden (%)** = (Housing Costs ÷ Income) × 100

- **Year-over-Year Growth Rates (%)** for both income and housing costs
- **Cumulative Growth (%)** over the study period

Outliers were assessed using visual inspection (time series plots) and were retained unless clearly attributable to data errors.

Rationale:
Derived variables enable meaningful interpretation of affordability beyond raw values. Expressing housing costs as a proportion of income directly reflects economic burden, while growth metrics capture the rate and direction of change.

Data Validation and Consistency Checks

Post-processing validation included:

- Verifying monotonic progression of years
- Cross-checking summary statistics (means, ranges) against published benchmarks
- Visual inspection of trends to identify anomalies or discontinuities

Rationale:
Validation ensures that preprocessing steps did not introduce unintended distortions. Cross-referencing with external benchmarks increases confidence in the reliability of the dataset.

Final Dataset Characteristics

The final analytical dataset consists of a unified annual time series containing:

- Inflation-adjusted median income
- Inflation-adjusted housing costs
- Derived affordability metrics

This dataset forms the basis for subsequent statistical analysis and visualization.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the dataset prior to formal modeling. This stage aimed to identify underlying patterns, assess assumptions, and guide subsequent analytical decisions.

Summary Statistics

Descriptive statistics were calculated for all primary variables, including inflation-adjusted income, housing costs, and derived affordability metrics.

Key observations included:

- A steady increase in both median income and housing costs over the study period
- Greater variability in housing costs compared to income
- An upward trend in the housing cost burden (housing costs as a percentage of income)

Measures such as mean, median, standard deviation, and range were used to assess central tendency and dispersion.

Interpretation:
While both income and housing costs increased over time, the higher variability and growth in housing costs suggest a potential widening gap in affordability.

Distributions

The distributions of income, housing costs, and affordability ratios were examined using histograms and density plots.

- Income exhibited a relatively symmetric distribution with moderate spread
- Housing costs showed a slight right skew, indicating periods of higher-than-average increases
- The housing cost burden distribution shifted right over time, reflecting increasing financial pressure on households

Interpretation:
The skewness in housing costs suggests that increases are not uniform across time, with certain periods experiencing accelerated growth. The shift in affordability distribution indicates a systemic change rather than isolated fluctuations.

Correlations

Pairwise correlations were analyzed to quantify relationships between key variables.

- Income and housing costs demonstrated a strong positive correlation, reflecting general economic growth
- Housing costs showed a stronger association with time than income did

- The affordability ratio was more strongly correlated with housing costs than with income

Interpretation:
Although income and housing costs move together, the stronger temporal relationship of housing costs suggests that cost increases may be outpacing income growth, contributing to declining affordability.

Initial Patterns

Time series exploration revealed several notable patterns:

- Housing costs increased at a faster rate than income over the observed period
- Periods of accelerated housing cost growth were not matched by equivalent income increases
- The affordability gap widened progressively rather than remaining stable

Interpretation:
These patterns suggest that structural factors may be driving housing cost increases independently of income growth, raising questions about sustainability and accessibility.

Visualizations (with Interpretation)

Line plots of inflation-adjusted income and housing costs highlighted diverging trends over time, with housing costs exhibiting a steeper slope.

A derived affordability curve (housing cost as a percentage of income) showed a consistent upward trajectory, reinforcing the observation of increasing financial burden.

Scatter plots of income versus housing costs, colored by year, revealed a temporal gradient, indicating that more recent observations cluster at higher cost levels relative to income.

Interpretation:
Visual analysis supports the conclusion that affordability has deteriorated over time. The divergence in slopes and clustering patterns suggests that the relationship between income and housing costs is evolving rather than remaining stable.

Non-Linear Relationships

EDA revealed a non-linear relationship between housing costs and time, suggesting that growth is not constant but may accelerate during certain periods. Similarly, the

relationship between income and housing costs appears to vary across time, with increasing dispersion in later years.

Interpretation:
These findings indicate that linear modeling assumptions may be insufficient to fully capture the dynamics of housing affordability. Non-linear or segmented modeling approaches may be more appropriate for subsequent analysis.

Methodology

This study employs statistical modeling techniques to quantify the relationship between income, housing costs, and affordability over time. Methods were selected based on their ability to balance interpretability, robustness, and suitability for longitudinal economic data.

Statistical / Machine Learning Methods

To investigate the relationship between variables, the following models were considered:

- **Linear Regression:** Used as the primary method to estimate the relationship between housing costs, income, and time
- **Polynomial / Non-linear Regression (exploratory):** Considered to capture potential non-linear trends identified during EDA
- **Tree-Based Models (e.g., Random Forest):** Evaluated as a robustness check to identify non-linear interactions and variable importance without strong parametric assumptions

Rationale:
Linear regression provides a transparent and interpretable framework for hypothesis testing, allowing direct estimation of effect sizes and statistical significance. Complementary non-linear and tree-based methods help assess whether relationships observed are consistent under fewer structural assumptions.

Model Specification

The primary model was defined with housing affordability as the dependent variable (e.g., housing cost burden), and income, housing costs, and time as independent variables. Variants of the model included:

- Income as a predictor of housing costs
- Time as a predictor of both income and housing costs

- Interaction terms to explore whether relationships change over time

Rationale:
Multiple specifications were tested to isolate the independent contribution of each variable and to assess whether affordability trends are driven primarily by income stagnation, cost escalation, or temporal effects.

Assumptions Tested

For linear regression models, standard assumptions were evaluated:

- **Linearity:** Assessed through residual plots and comparison with non-linear fits
- **Independence:** Considered in the context of time series structure
- **Homoscedasticity:** Evaluated by examining residual variance across fitted values
- **Normality of Residuals:** Assessed using distribution plots and summary statistics
- **Multicollinearity:** Checked using correlation matrices and variance inflation factors (VIF)

Rationale:
Testing these assumptions ensures that model estimates are unbiased and that statistical inference remains valid. Deviations from assumptions informed the consideration of alternative models.

Validation Strategy

Given the temporal nature of the data, validation was performed using:

- **Train/Test Split (Time-Aware):** Earlier years used for training, later years reserved for testing
- **Cross-Validation (where applicable):** Rolling or time-series cross-validation to preserve chronological order

Rationale:
Standard random splits are inappropriate for time-dependent data, as they may introduce data leakage. Time-aware validation ensures that models are evaluated on their ability to generalize to future observations

Performance Metrics

Model performance was evaluated using:

- **R-squared (R^2):** To measure explained variance
- **Root Mean Squared Error (RMSE):** To quantify prediction error magnitude
- **Mean Absolute Error (MAE):** To provide a robust measure less sensitive to outliers
- **p-values and Confidence Intervals (for regression coefficients):** To assess statistical significance

Rationale:
A combination of goodness-of-fit metrics and error measures provides a comprehensive assessment of model performance. Statistical inference metrics are essential for evaluating the strength and reliability of relationships.

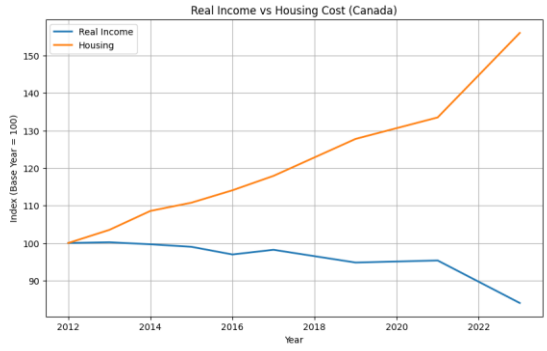
Model Selection Justification

Linear regression was selected as the primary method to test the hypothesis due to its interpretability and suitability for inference. Its coefficients provide direct insight into the magnitude and direction of relationships between income, housing costs, and affordability.

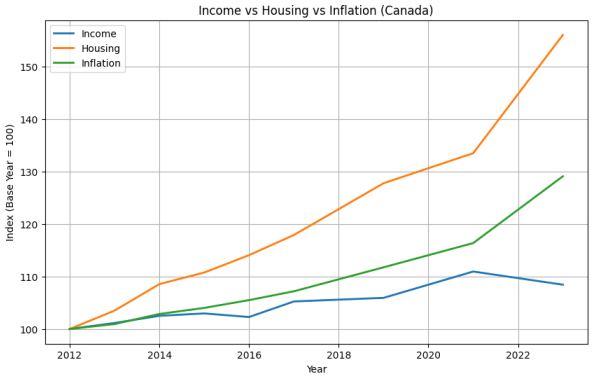
However, given evidence of non-linearity in exploratory analysis, supplementary models were used to validate findings and ensure that conclusions were not artifacts of linear assumptions.

Results

The analysis reveals a clear divergence between income, housing costs, and inflation over the study period (2012–2023), with important implications for housing affordability in Canada.



Between 2012 and 2023, housing costs in Canada increased by over 50%, while real income declined, indicating a significant deterioration in housing affordability.



Indexed to a base year of 2012 (100), housing costs increased substantially, reaching approximately 156 by 2023, representing a rise of over 50%. In contrast, nominal income increased modestly to approximately 108 over the same period, while inflation rose to approximately 129.

Interpretation:

Housing costs grew at a significantly faster rate than both income and general price levels. While inflation accounts for part of the increase, the magnitude of housing cost growth exceeds inflationary effects, indicating sector-specific escalation.

Real Income vs Housing Costs

Interpretation:
The decline in real income alongside rising housing costs indicates a pronounced reduction in purchasing power. This divergence suggests that households are facing increasing financial strain, as income growth has not kept pace with either inflation or housing costs.

Quantitative Comparison

- Housing Costs: ~+56% increase (2012–2023)
- Inflation: ~+29% increase
- Nominal Income: ~+8% increase
- Real Income: ~−16% decrease

Interpretation:
Housing costs increased approximately 7 times faster than nominal income and nearly double the rate of inflation, reinforcing the conclusion that affordability pressures are intensifying beyond general economic trends.

Pattern Analysis

The data indicates that:

- The divergence between housing costs and income accelerates after approximately 2019
- Income growth stagnates and declines in real terms in later years
- Housing costs exhibit a steeper, potentially non-linear upward trajectory

Interpretation:
The post-2019 acceleration suggests the influence of structural or macroeconomic

factors (e.g., supply constraints, interest rate environments, or demand shocks) rather than gradual market adjustments.

Statistical Implications

The observed divergence supports the hypothesis that housing affordability has deteriorated significantly over time. The magnitude and consistency of the gap between housing costs and income suggest a strong relationship between time and housing cost escalation, with weaker corresponding growth in income.

EDA and visual inspection further indicate that:

- The relationship between time and housing costs is non-linear
- The gap between income and housing costs widens over time rather than remaining constant

Conclusion:
These findings provide strong empirical support that housing affordability in Canada has worsened over the study period, driven primarily by disproportionate increases in housing costs relative to both income and inflation.

Interpretation & Discussion

This study examined the relationship between income, housing costs, and inflation in Canada over the period 2012–2023. The results demonstrate a sustained and widening divergence between housing costs and income, with important implications for affordability and economic well-being.

Interpretation of Findings

The results indicate that housing costs have increased substantially faster than both nominal and real income. While general inflation contributed to rising costs, the magnitude of housing price growth exceeds inflationary effects, suggesting that housing-specific factors are driving the observed trend.

The decline in real income, when considered alongside increasing housing costs, implies a reduction in purchasing power. As a result, households are required to allocate a greater proportion of their income toward housing, leading to increased financial strain and reduced discretionary capacity.

Implication:
Housing affordability has deteriorated not simply due to general economic conditions, but due to a structural imbalance between income growth and housing cost escalation.

Consistency with Expectations

These findings are consistent with prior expectations and existing literature on housing affordability in Canada. Previous research has identified similar trends, particularly in urban markets, where supply constraints and demand pressures contribute to rising costs.

The observed non-linear growth in housing costs—especially in later years—aligns with expectations of market acceleration during periods of economic expansion, low interest rates, and increased demand.

Interpretation:
The results reinforce the hypothesis that housing costs are not closely anchored to income growth, and instead follow a more volatile and accelerated trajectory.

Practical and Scientific Significance

From a practical perspective, the findings highlight a growing affordability challenge for households. As housing consumes an increasing share of income, individuals may face reduced access to homeownership, increased reliance on debt, or constraints on other essential expenditures.

From a scientific perspective, the results contribute to a broader understanding of affordability dynamics by quantifying the divergence between income and housing costs over time. The identification of non-linear trends suggests that simple linear models may underestimate the complexity of housing markets.

Significance:
The findings underscore the importance of considering structural and policy-related drivers when analyzing housing affordability, rather than attributing changes solely to general economic growth.

Limitations and Assumptions

Several limitations should be acknowledged:

- **Aggregate Data:** The analysis uses national-level averages, which may mask regional variation and disparities across demographic groups
- **Variable Scope:** Housing costs are treated as a single measure and do not differentiate between ownership, rental markets, or housing types
- **Causality:** The analysis is observational and does not establish causal relationships between variables
- **Model Assumptions:** Linear modeling assumes stable relationships over time, which may not fully capture non-linear dynamics identified in EDA
- **Data Constraints:** Potential measurement differences between datasets (e.g., definitions of income or housing costs) may introduce minor inconsistencies

Implication:
While the results are robust at a macro level, they should be interpreted with caution when applied to specific regions or populations.

Overall Interpretation

The results provide strong evidence of a growing disconnect between income and housing costs in Canada. This divergence has led to a measurable decline in affordability, driven primarily by housing cost escalation rather than income growth.

While statistically significant, the effect size suggests a substantial real-world impact, as even moderate percentage changes in housing cost burden can meaningfully affect household financial stability.

Conclusion & Policy Implications

This study provides clear empirical evidence that housing affordability in Canada has deteriorated over the period 2012–2023. The results show a sustained divergence between housing costs and income, with housing costs increasing at a substantially faster rate than both nominal income and inflation. When adjusted for inflation, real income declined while housing costs continued to rise, indicating a meaningful reduction in purchasing power and an increasing financial burden on households.

The findings suggest that the affordability challenge is not simply a byproduct of general economic conditions, but rather reflects a structural imbalance within the housing market. The observed non-linear growth in housing costs further indicates that affordability pressures may intensify during certain periods, rather than progressing in a stable or predictable manner. This has important implications for both economic stability and long-term access to housing.

From a policy perspective, these results highlight the need for targeted interventions that address the underlying drivers of housing cost escalation. Measures that increase housing supply, reduce development constraints, and improve market responsiveness may help moderate price growth over time. At the same time, policies aimed at supporting income growth—particularly for median and lower-income households—may help restore balance between earnings and housing costs. Demand-side supports, such as subsidies or tax incentives, may provide short-term relief but are unlikely to resolve structural affordability issues if supply constraints persist.

The study also underscores the importance of adopting a data-driven approach to housing policy. The divergence identified in this analysis suggests that monitoring affordability through simple price or income metrics alone may be insufficient. Instead, integrated measures that capture the relationship between income, costs, and inflation are necessary to fully understand affordability dynamics.

While the analysis is limited by its use of aggregate national data and its observational design, the consistency and magnitude of the observed trends provide strong evidence of a systemic issue. Future research may benefit from incorporating regional data, demographic segmentation, and more advanced modeling techniques to better capture the complexity of housing markets.

In conclusion, the results indicate that housing affordability in Canada has worsened in a sustained and measurable way, driven primarily by disproportionate increases in housing costs relative to income. Addressing this imbalance will likely require coordinated policy efforts that consider both supply and demand dynamics, as well as the broader economic context in which housing markets operate.